On the Conservation Laws and Pathway Realizability of Biochemical Reaction Networks

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Abstract

Background: Conserved metabolite pools and substrate availability constrain biochemical pathways before kinetic parameters are known. These properties are usually evaluated on incompatible mathematical abstractions: stoichiometric nullspaces omit shared catalytic participants, while graph reachability ignores token multiplicities. Because participants that appear identically on both reaction sides are eliminated under net stoichiometry despite being required for firing, standard reductions can destroy catalytic gating and alter deficiency.
Results: We present the chemical reaction network (CRN) submodule of `SynKit`, an open-source Python toolkit with an optimized C++ core. It builds networks from curated `KEGG` modules, structural `SBML` models, and generative reaction rules into a unified bipartite representation preserving distinct reactant and product incidences. On this representation, `SynKit` computes exact conservation laws, minimal semiflows and siphons, deficiency, and discrete firing certificates that verify whether a reaction multiset can execute from a finite inventory. Twelve reference networks match ground-truth values and agree with `CRNT4SBML` across eight structural invariants, whereas floating-point bases yield corrupted conservation vectors in three cases. Ten `BioModels` entries, up to 504 species and 827 reactions, pass structural round trips. In generative chemistry, formose expansion across six depths yields identical sets of 10 572 species and 11 239 reactions across `SynKit`, `MØD`, and `MolFoundry`. In `KEGG` glycolysis, net stoichiometric reduction of shared polyphosphate preserves the net matrix but lowers deficiency from one to zero and falsely permits firing without cofactors. A 15-step pathway certificate realizes two pyruvates from glucose when currency metabolites are omitted, while exhaustive search proves the complete 24-species network is unrealizable without initial cofactor replenishment. In the formose network, the single conservation law reproduces carbon counts for all 37 species, yet precursor pools sharing identical six-carbon budgets diverge in their ability to supply tetrose and hexose targets.
Conclusions: Stoichiometric conservation is necessary but insufficient to guarantee pathway execution. By unifying conservation analysis, deficiency invariants, and Petri-net reachability on a single incidence representation with exact rational arithmetic, `SynKit` makes participant retention, cofactor boundaries, and precursor constraints explicit and testable. Source code, recorded benchmarks, and reproducible analysis workflows accompany the manuscript.

Keywords: Chemical reaction networks, chemical reaction network theory, Petri nets, conserved moieties, stoichiometry, pathway realizability, `SBML`

1 Background

Curated databases such as `KEGG` [1], `Reactome` [2], and `WikiPathways` [3] catalog known metabolic pathways and biochemical reactions, while graph-transformation methods enumerate candidate reactions from molecular seeds and transformation rules [4]. Before assigning kinetic parameters, structural analyses clarify the behavioral constraints imposed by network topology. Specifically, researchers examine which metabolite

pools are conserved by stoichiometric invariants [5], which qualitative dynamical bounds or multistationarity criteria are governed by reaction network structure [6, 7], and whether a candidate pathway is realizable from a finite substrate inventory [8].

Assessing these constraints simultaneously is difficult because each analysis relies on a different mathematical abstraction of the biochemical system. A species graph captures connectivity between metabolites but loses the joint requirement of multiple co-substrates [9]. The net stoichiometric matrix records net changes in species abundance but omits participants that appear identically as reactants and products, such as catalysts or recycled cofactors. For example, a catalyzed reaction $A + C \rightarrow B + C$ and the uncatalyzed conversion $A \rightarrow B$ yield identical net columns in $\mathbf{S}$, yet only the first requires the catalyst C to proceed, and the two define different reaction complexes that alter network deficiency. Similarly, a balanced reaction combination or a qualitatively reachable product does not establish that the reactions can proceed in an admissible sequence from a finite inventory [8]. Representation choices during model import can therefore compromise the assumptions of another analysis.

Because of these distinct formulations, existing software packages specialize in separate subsets of analyses: `COBRApy` [10] supports constraint-based flux analysis [11], `CRNT4SBML` [12] evaluates deficiency and potential bistability, `MØD` [4] explores rule-generated reaction spaces, `Tellurium` [13] and `libRoadRunner` [14] simulate kinetic ODEs, and Petri-net methods analyze metabolic invariants [15, 16] and provide pathway realizability certificates [8]. Standard exchange formats such as `SBML` [17] and `COMBINE` [18], alongside converters like `KEGGtranslator` [19], facilitate model transfer. However, converting models between separate formats can inadvertently discard structural features, such as catalysts or boundary species, that specific analyses require.

Conserved pools and substrate sufficiency therefore need to be examined on the same underlying reaction model. In glycolysis [1], a route appears connected from glucose to pyruvate when cofactors are omitted, yet cannot proceed when ATP and redox cofactors are retained with finite initial pools. In a rule-generated formose network [20], every reaction may conserve carbon, yet a proposed pathway remains unrealizable if its reactions require more glycolaldehyde molecules than the initial precursor pool provides. In both cases, conservation laws alone cannot establish whether a reaction sequence is realizable from a specified finite inventory [8].

Here, we present the chemical reaction network (CRN) submodule of `SynKit` [21] to evaluate these constraints on a shared representation. By connecting `KEGG` pathway import, structural `SBML` exchange, and rule-based generation to exact conservation laws, deficiency theory, and Petri-net reachability, the toolkit tests whether reaction sequences can execute without discarding catalysts or cofactors. We validate the implementation across twelve reference networks against `CRNT4SBML`, verify structural round trips on ten `BioModels` entries, and benchmark exact enumeration limits. Case studies in glycolysis and formose demonstrate how cofactor boundaries, participant retention, and precursor mixtures alter deficiency and pathway realizability even when net stoichiometry remains unchanged.

2 Implementation

2.1 Network representation and participant incidence

`SynCRN` serves as the core reaction-network data structure in `SynKit`, maintaining species, reactions, transformation rules, and chemical annotations (Fig. 1). It derives bipartite digraphs in `NetworkX` [22], stoichiometric matrices, and Petri nets for downstream structural analysis. The bipartite digraph has species vertices X and reaction vertices $\mathscr{R}$, and each reaction vertex is linked to its reactants and products. It is therefore the incidence representation of a directed hypergraph in which every reaction is a hyperedge, so multi-substrate requirements remain joint hyperedge tails rather than pairwise species edges.

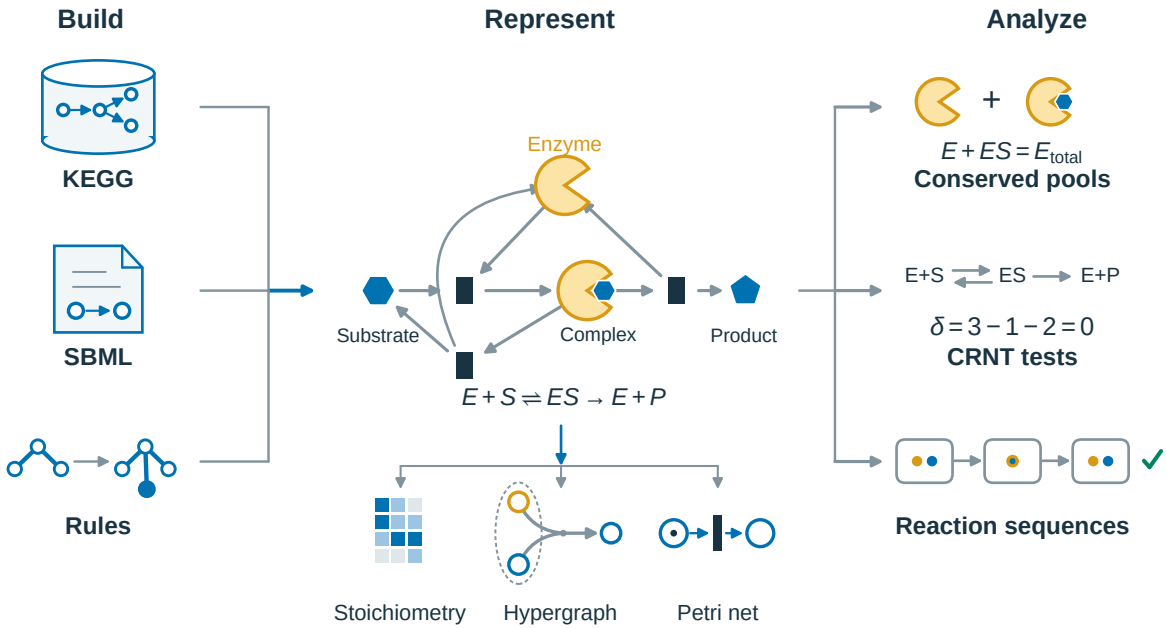


Fig. 1: Overview of `SynKit CRN` workflows across construction, representation, and structural analysis. **Build:** Reaction networks imported from curated databases (`KEGG`), exchange files (`SBML`), or generated by rule expansion are mapped to a unified data structure. **Represent:** An illustrative enzyme mechanism ($E+S \rightleftharpoons ES \rightarrow E+P$) is converted into three complementary views: stoichiometric matrices, directed hypergraphs, and Petri nets, ensuring that multi-reactant requirements (E and S) are captured as joint inputs. **Analyze:** These structural representations enable exact conservation analysis (identifying the conserved pool $E + ES$), chemical reaction network deficiency calculation ($\delta = 0$), and discrete pathway realizability through verified firing sequences. Blue circles denote metabolites, orange circles denote enzymes, and dark bars represent reactions.

For reaction r, reactant and product multisets have coefficients s^-_{xr} and s^+_{xr}, defining separate reactant and product (pre/post) incidence matrices $\mathbf{S}^- = (s^-_{xr})$ and $\mathbf{S}^+ = (s^+_{xr})$ alongside the net stoichiometry $\mathbf{S} = \mathbf{S}^+ - \mathbf{S}^-$. Retaining participants present on both sides in both matrices preserves catalytic and cofactor requirements. Their net coefficients cancel in $\mathbf{S}$, but they still govern reaction enabling and define distinct reaction complexes (Fig. 2a,b). In Petri-net terms, such shared species form self-loops, and the net incidence matrix alone cannot determine whether transitions are enabled [16]. Construction does not establish elemental balance automatically, so chemical interpretation requires suitable input equations and

annotations. The conditions under which a reaction network represents genuine chemical transformations are themselves a subject of study [23].

Reaction networks can be constructed from curated databases, generative chemistry, or standard model files. The `Query` module parses reaction equations from `KEGG`, `Construct` expands networks from chemical seeds using reaction rules, and `IO` imports models from `SBML`. During construction, reversible reactions can be partitioned into directed steps. Users can also choose whether to retain cofactors (such as ATP and NADH) and boundary species, which determines the resulting conserved moieties, network deficiency, and discrete pathway realizability.

2.2 Conserved moiety and stoichiometric analysis

The stoichiometric matrix $\mathbf{S}$ records the net change in every species for each reaction. For concentrations c and reaction rates $v(c)$, the dynamics are written as $\dot{c} = \mathbf{S}v(c)$. A vector y satisfying $y^T\mathbf{S} = 0$ defines a conserved linear quantity $y^T c$, because the reactions can redistribute the participating species without changing this weighted total. `SynKit CRN` computes a rational basis for the left nullspace of $\mathbf{S}$ and converts its vectors to primitive integer representatives. These vectors describe all rational conservation relations, but they need not be non-negative or identify the smallest conserved pools.

Minimal P-semiflows are computed separately by exact Farkas elimination [24]. They are non-zero, non-negative integer vectors in the left kernel of $\mathbf{S}$ with inclusion-minimal supports, normalized by their greatest common divisor. Their supports identify minimal conserved pools of the represented network [5]. A P-semiflow guarantees mathematical conservation, whereas assigning its support to a chemical moiety requires species annotations and biochemical interpretation.

T-semiflows satisfy $\mathbf{S}x = 0$ with $x \geq 0$ and describe balanced combinations of reactions whose total net species change is zero. They are reaction-space relations, not conserved species pools (Fig. 2c). A T-semiflow fixes neither a firing order nor an initial marking, and it does not imply thermodynamic feasibility. All-positive left and right kernel vectors define conservativity and consistency, respectively. To determine positive-kernel feasibility, a linear program proposes a candidate vector that is accepted only after exact rational verification. Feasible networks are certified by an exact all-positive vector, infeasible networks by a rational separating vector, and unresolved solver outcomes remain undecided. Numerical kernel and Jacobian-pattern checks can also characterize approximate subspaces and qualitative matrix patterns, but because floating-point estimates do not isolate integer minimal supports, numerical rounding cannot substitute for exact conservation analysis.

2.3 Deficiency analysis and multistationarity bounds

Complexes are distinct reactant or product multisets. With n complexes, ℓ connected components of the complex graph (linkage classes), and stoichiometric rank s, deficiency is $\delta = n - \ell - s$. A network is weakly reversible when every linkage class is strongly connected. `SynKit CRN` calculates ranks exactly over $\mathbb{Q}$ and

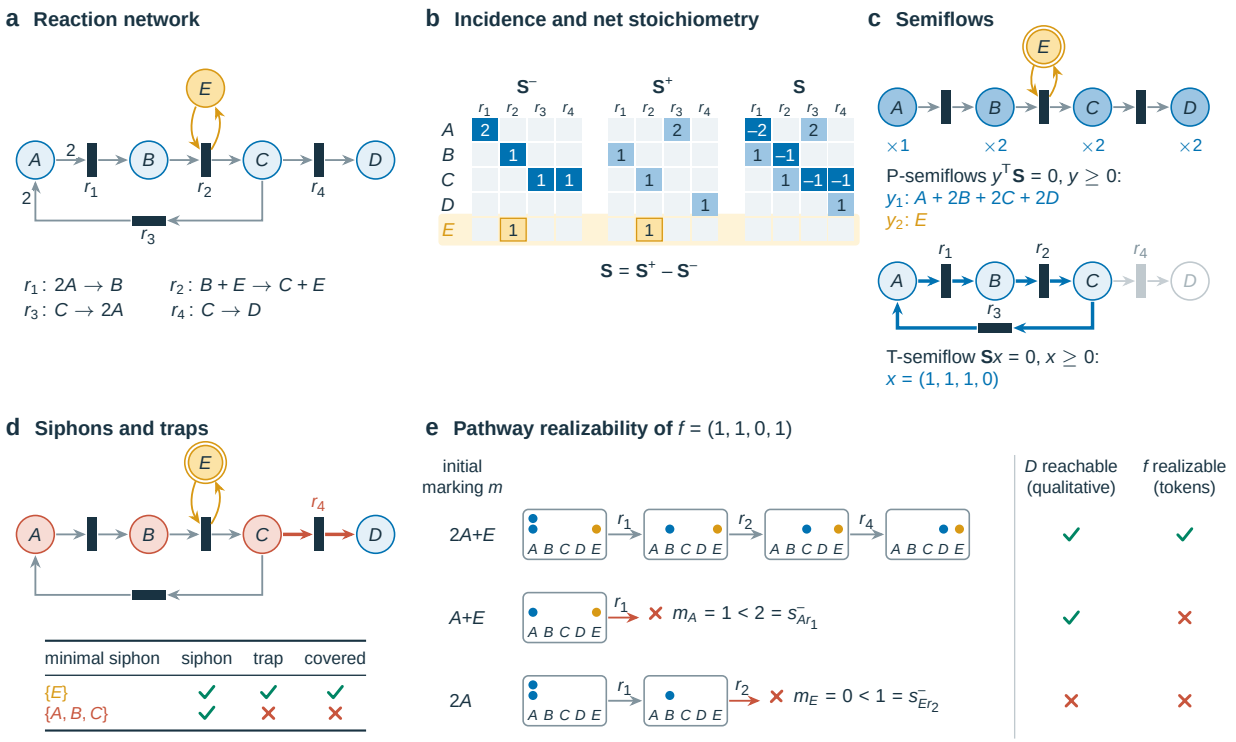


Fig. 2: Structural objects on one toy network, colored as in Fig. 1, with red marking blocked steps or uncovered sets. **(a)** Petri net (circles: species, bars: reactions, arc labels: coefficients other than one). The enzyme E is consumed and returned by r_2. **(b)** Reactant, product, and net incidence matrices. The row of E cancels in $\mathbf{S}$, so the net matrix alone does not show that r_2 requires E. **(c)** P-semiflows (shaded, with weights) are conserved pools. The T-semiflow (heavy arcs) is a balanced reaction combination that fixes no firing order. **(d)** Minimal siphons. $\{E\}$ contains a conserved pool and is covered, whereas $\{A, B, C\}$ drains through r_4 (heavy) and is uncovered. **(e)** Firing from three initial markings (dots: tokens). From $A + E$, qualitative reachability reaches D although no reaction is enabled. Inequalities give the failed condition $m_x \geq s^-_{xr}$.

reports the deficiency of the network and of each linkage class, weak reversibility, and the terminal strong linkage classes.

For positive mass-action rate constants, a weakly reversible deficiency-zero network has exactly one positive steady state in each positive stoichiometric compatibility class $(c_0 + \operatorname{im}\mathbf{S}) \cap \mathbb{R}^X_{>0}$, and it is locally asymptotically stable relative to that class [25, 6]. A deficiency-zero network that is not weakly reversible has no positive steady state. The Deficiency One Theorem gives at most one positive steady state per positive stoichiometric compatibility class when each linkage-class deficiency is at most one, their sum equals δ, and each linkage class has one terminal strong linkage class, while weak reversibility additionally ensures existence [26]. Failure of these hypotheses leaves the corresponding theorem undecided. Introductions to this body of theory for biologists are available [7]. Criteria that reach beyond deficiency, including injectivity [27], concordance [28], and graph-theoretic multistationarity tests [29], are not implemented here, and a network left undecided by the deficiency theorems is not thereby shown to be multistationary.

2.4 Pathway realizability and siphon analysis

A siphon is a non-empty species set for which every reaction producing a member also consumes a member. Once empty, it cannot refill under Petri-net firing semantics. A trap has the dual property that every reaction consuming a member also produces a member, so a marked trap cannot become empty [16]. `SynKit CRN` enumerates minimal siphons using implication propagation and closure minimization, within the branch-and-bound approach to structural enumeration [30]. Producing a selected species forces selection of a reactant. Selected and excluded species are propagated before branching. Each reported minimal set excludes its supersets from further search, and a maximal-closure deletion test verifies minimality. Traps use the dual incidence (Fig. 2d). This prunes many subsets but does not remove worst-case combinatorial growth.

The package checks whether every minimal siphon contains the support of a P-semiflow, and we call such a siphon covered. Under the boundedness and kinetic assumptions of the corresponding persistence result, this provides a sufficient persistence certificate for positive initial conditions [31]. An uncovered siphon means the test is insufficient, and does not prove extinction or identify a unique nutrient supplementation strategy. The returned details expose these uncovered sets.

A marking m specifies the non-negative integer inventory of each species. Reaction r is enabled precisely when $m_x \geq s_{xr}^-$ for every species, and firing gives $m' = m + \mathbf{S}_{\cdot r}$. Qualitative reachability instead accumulates reachable species sets and ignores consumption and multiplicity. Given a requested flux $f \in \mathbb{N}_0^{|\mathscr{R}|}$, realizability search seeks an enabled sequence firing each reaction exactly f_r times [8]. A positive result includes a certificate consisting of an ordered reaction list that can be checked by updating the inventory after every step. An exhaustive negative result is `unrealizable`, while reaching a state or depth limit is `inconclusive`. Fig. 2e contrasts the two notions on one requested flux. Certificates establish firing in the specified discrete model only.

2.5 Implementation and software comparison

The software employs a hybrid architecture where users interact entirely through a high-level Python interface, while computationally intensive tasks are transparently dispatched to an optimized C++17 library. These native routines compute exact rational nullspaces, CRNT deficiency invariants, minimal semiflows, siphon enumeration, and discrete firing sequences. To eliminate floating-point rounding errors and ensure exact certificates, linear-algebraic and combinatorial routines use arbitrary-precision integer arithmetic via Boost.Multiprecision. For generative chemistry, `SynReactor` applies double-pushout graph transformations with implicit hydrogen counts. Its native C++ controller performs candidate construction, frontier updates, matching, rewriting, and reaction integration from seed molecules. Following the `MØD` strategy framework [32], each rule application must use at least one newly generated molecule.

To ensure interoperability with the wider modeling community, `SynKit` provides native `SBML` Level 3 Core import and export. The parser preserves species identifiers, compartments, and exact stoichiometries across export and re-import, omitting kinetic rate laws and package extensions. Exported models provide

the structural foundation for parameterization in kinetic simulators like `Tellurium` or constraint-based packages like `COBRApy`.

Through this design, `SynKit` occupies a distinct niche in the systems biology software ecosystem (Table 1). Established packages typically specialize in continuous kinetic simulation (`Tellurium`), flux-balance analysis (`COBRApy`), or algebraic deficiency testing (`CRNT4SBML`). While `crnpy` evaluates exact semiflows and `MolFoundry` solves rule-based hyperflows, `SynKit` unifies rule generation, deficiency, minimal siphons, and discrete firing certificates on a shared model. Downstream analyses can then build on exported `SBML` networks. Section 1 of the Supplementary Information and Listing S1 provide complete implementation details and usage examples.

Table 1: Feature comparison of biochemical reaction-network software (✓ supported, (✓) partial or indirect as detailed in footnotes, × not supported). Capabilities are determined from official documentation and source code. `CRNT4SBML` is experimentally evaluated on shared benchmark networks (Table 2), while formose generation is compared with `MØD` and `MolFoundry`, and fixed-route realizability with `MolFoundry` (Section 3.5). The `SynKit` column describes its chemical reaction network (CRN) submodule.

	`SynKit` (this work)	`CRNT4SBML` [12]	`CoNtRol` [33]	`crnpy` [34]	`COBRApy` [10]	`MØD` [4]	`MolFoundry` [35]	`Tellurium` [13, 14]
Network construction and exchange								
`KEGG` pathway import	✓	×	×	×	×	×	×	×
`SBML` import	(✓)[a]	✓	×	✓	✓	×	×	✓
`SBML` export	(✓)[a]	×	×	✓	✓	×	✓	✓
Rule-based network generation	✓	×	×	×	×	✓	✓	×
Structural analysis								
Deficiency and weak reversibility (CRNT)	✓	✓	✓	✓	×	×	×	×
Conservation laws (left-nullspace basis)	✓	✓	×	✓	×	×	×	✓
Minimal semi-positive conservation laws	✓	×	×	✓	×	×	×	×
Elementary flux modes or T-semiflows	✓	×	×	✓	×	×	×	×
Minimal siphons and traps	✓	×	(✓)[b]	×	×	×	×	×
Dynamics and pathway feasibility								
Multistationarity analysis	×	✓	✓	(✓)[c]	×	×	×	×
Pathway realizability (firing sequence)	✓	×	×	×	×	(✓)[d]	✓[f]	×
Constraint-based modeling (e.g., FBA)	×	×	×	×	✓	×	(✓)[f]	×
Kinetic simulation	×	(✓)[e]	×	×	×	×	✓[g]	✓

[a] `SBML` Level 3 Version 2 Core reaction structure only, with kinetic laws and rules not handled.
[b] Siphons underlie persistence tests, but minimal-siphon enumeration is not exposed as an output.
[c] `crnpy` supplies Gröbner-basis, elementary-mode, and ACR machinery rather than bistability search.
[d] `MØD` evaluates integer hyperflows rather than discrete firing certificates over a finite marking.
[e] ODE simulation is provided in support of `CRNT4SBML`'s bistability analysis.
[f] `MolFoundry` solves integer hyperflows over derivation graphs (Gurobi or SMT) for firing orders.
[g] Rule-based stochastic simulation with on-demand network expansion.

2.6 Validation and benchmark design

We validate the toolkit through analytical benchmarks, runtime scaling experiments, and biochemical case studies. First, structural correctness is evaluated across twelve reference networks spanning enzyme kinetics, phosphorylation cycles, bistable switches, and motifs exhibiting concentration robustness, with shared CRNT invariants independently confirmed by `CRNT4SBML`. Second, computational scaling for exact siphon and conserved-pool enumeration is profiled across synthetic topologies to establish practical tractability limits. Third, `SBML` model exchange and deficiency preservation are tested across eleven curated `BioModels` models up to 504 species and 827 directed reactions.

Two mechanistic case studies demonstrate how structural representation choices dictate pathway feasibility from finite metabolite pools. In `KEGG` glycolysis, we assess how retaining catalytic polyphosphate and closing adenine/nicotinamide cofactor cycles govern discrete pathway execution. In rule-generated formose chemistry, we contrast qualitative hypergraph reachability with stoichiometric realizability under finite precursor budgets. Each feasible pathway is confirmed by an ordered reaction sequence from available precursors, while infeasibility is proved through exhaustive search.

3 Results and discussion

3.1 Reference-network validation and comparison with `CRNT4SBML`

All twelve reference networks match their analytical and algebraic ground-truth values (Table 2, with sources and validation objectives in Table S1). The benchmark suite spans classic enzyme kinetics (Michaelis–Menten and its reversible counterpart), multisite phosphorylation cycles [36], the bistable Edelstein switch [37], and the Shinar–Feinberg robustness motif [38]. Constructed synthetic cases also verify boundary conditions. Disconnected reversible pairs confirm that independent conserved pools are correctly isolated without conflating disjoint subnetworks, while the open reactor ($0 \to A \to B \to 0$) tests exact handling of inflow and outflow through the zero complex.

Table 2: Structural invariants of the twelve reference networks: species, directed reactions, complexes (n), linkage classes (ℓ), stoichiometric rank (s), deficiency (δ), conservation dimension (laws), and weak reversibility (WR). Every network matches its analytical and algebraic ground truth, and `CRNT4SBML` reproduces all eight shared structural quantities.

Network	Species	Reactions	n	ℓ	s	δ	Laws	WR
Reversible isomerization	2	2	2	1	1	0	1	yes
Irreversible isomerization	2	1	2	1	1	0	1	no
Isomerization cycle	3	3	3	1	2	0	1	yes
Disconnected reversible pairs	6	6	6	3	3	0	3	yes
Pairs with stoichiometry	6	6	6	3	3	0	3	yes
Michaelis–Menten	4	3	3	1	2	0	2	no
Reversible Michaelis–Menten	4	4	3	1	2	0	2	yes
Edelstein switch	3	6	5	2	2	1	1	yes
Phosphorylation: one site	6	6	6	2	3	1	3	no
Phosphorylation: two sites	9	12	10	2	6	2	3	no
Shinar–Feinberg motif	2	2	4	2	1	1	1	no
Open reactor	2	3	3	1	2	0	0	yes

To benchmark structural interoperability, `CRNT4SBML` 0.0.15 independently parsed all twelve exported `SBML` models. Across all twelve benchmarks, both tools yielded identical results for all eight shared structural quantities (Table 2), confirming topological consistency across independent implementations. Standard floating-point nullspace methods produce numerical residuals on all closed networks, where scaling and rounding fail to recover valid integer conservation vectors on three of the twelve models (Table S1). Exact rational arithmetic guarantees that the conservation identity $y^T\mathbf{S} = 0$ holds across all networks without rounding error.

3.2 Computational cost of exact structural enumeration

Enumerating minimal siphons (Fig. 2d) and exact conserved pools (Fig. 2c) identifies extinction-prone species sets and mass conservation laws, but naive search scales exponentially with $2^N - 1$ candidate subsets. To prevent combinatorial explosion, implication pruning propagates stoichiometric constraints along active reactions during search (Fig. 3a). In reversible chains, selecting an initial species forces each adjacent neighbor, isolating the entire chain as the unique minimal siphon without evaluating spurious sub-combinations. Across five repeats per size, pruned search returns identical siphon sets with a 46.6-fold speedup at 21 species (0.63 ms versus 29.25 ms for exhaustive search, Fig. 3b).

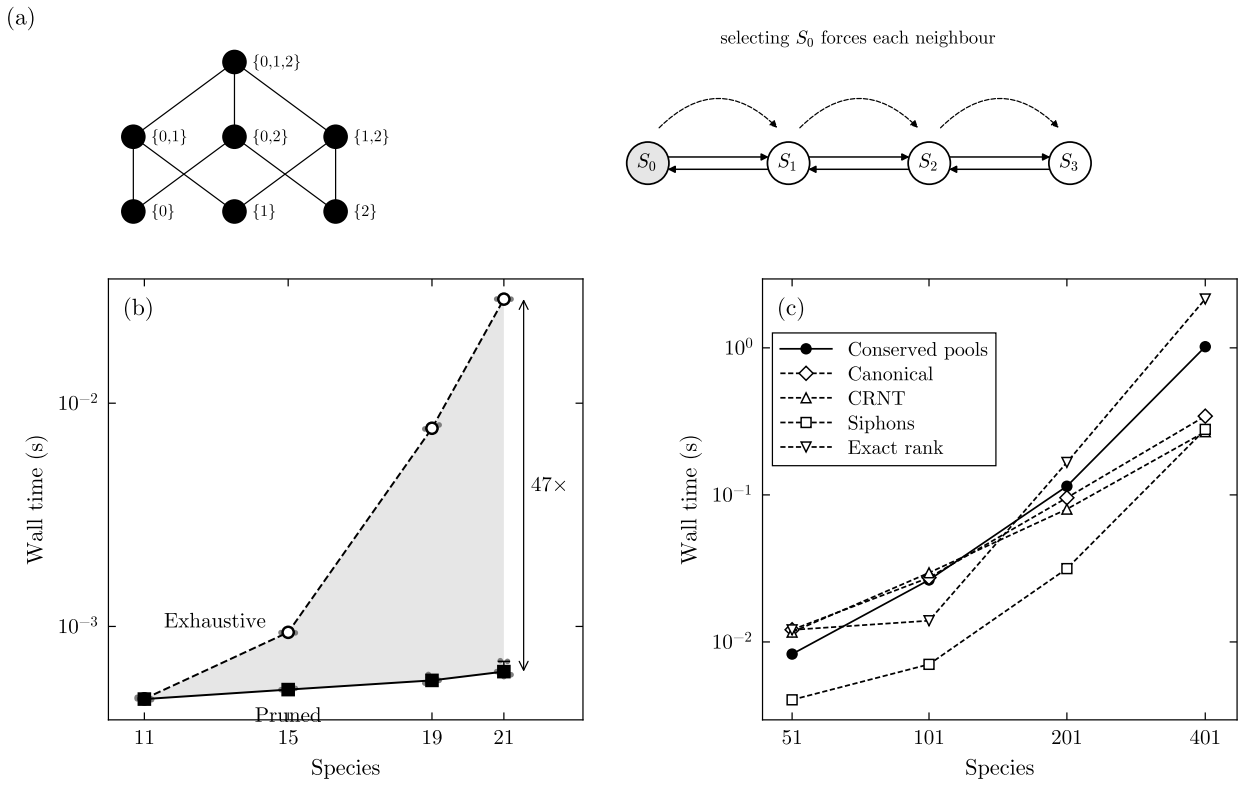


Fig. 3: Search algorithms and runtime scaling on reversible-chain networks. **(a)** Schematic of exhaustive power-set search versus implication pruning, where selecting S_0 forces all chain species into the minimal siphon. **(b)** Minimal-siphon runtimes across five repeats per size (points: individual runs, markers: medians and IQRs, bracket: speedup ratio at 21 species). **(c)** Wall times for five structural analyses across chains up to 401 species.

The broader benchmark suite evaluates scalability across directed chains, reversible chains, and random sparse bimolecular networks (Fig. S3). All 60 requested configurations complete without numerical failure. On the 401-species reversible chain, minimal siphons (Fig. 2d) take 0.279 s, canonicalization 0.343 s, exact conserved pools (Fig. 2c) 1.02 s, and standalone exact rank 2.15 s (Fig. 3c). The random networks with 49, 99, 197, and 395 species contain 331, 272, 47, and 87 minimal siphons, completing siphon and conserved-pool enumeration in 0.113 s and 0.0789 s at 395 species. Stoichiometric implication pruning and rational nullspace methods thus keep exact structural enumeration computationally practical on networks with hundreds of species.

3.3 Structural SBML round-trip validation in BioModels

To test real-world structural interoperability and scalability, we evaluated eleven curated BioModels [39] entries spanning diverse sizes, including ten reaction networks and one qualitative network control (SBML-qual). All ten reaction-bearing models successfully round-trip through SynKit's SBML parser and exporter without loss of stoichiometry, and their deficiency values match the analytical identities (details in Table S2). Even on the largest model (BIOMD0000000255, with 504 species and 827 directed reactions), SynKit identifies 75 minimal siphons in 0.0235 s and 17 exact conserved pools in 2.15 s (Fig. 4a). Testing siphon coverage by conserved pools evaluates whether species are structurally protected from extinction. In BIOMD0000000255 and the 100-species model BIOMD0000000019 (20 siphons, 19 pools), at least one minimal siphon lacks a supporting conserved pool, indicating that these cascades contain components that can drain to extinction unless sustained by continuous replenishment. The qualitative control BIOMD0000000562 is correctly identified as reaction-free for Core structural analysis.

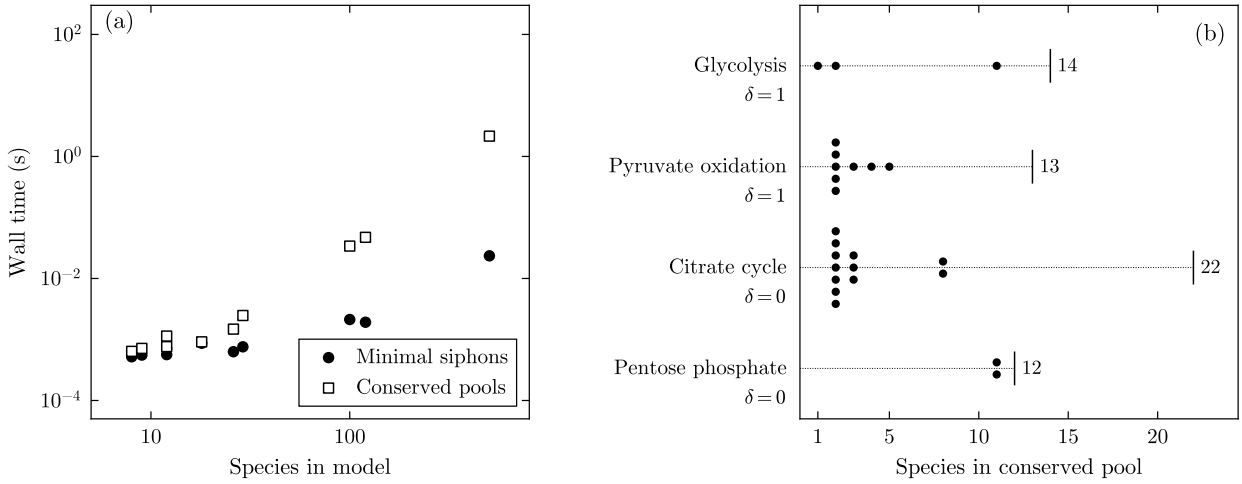


Fig. 4: Enumeration runtimes in BioModels and conserved pools in KEGG modules. **(a)** Wall time for completed minimal-siphon and exact conserved-pool enumeration in ten reaction-bearing BioModels models. All ten models complete both enumerations. **(b)** Minimal non-negative conserved pools in four reduced KEGG modules. Each dot is one pool placed at its number of species, pools of equal size are stacked, and the dotted line spans the module species, ending at the module size.

3.4 Conservation and cofactor requirements in glycolysis

Four KEGG [1] pathway modules illustrate structural conservation across metabolic subsystems: glycolysis (M00001), pyruvate oxidation (M00307), the citrate cycle (M00009), and the pentose phosphate pathway (M00004). Their conserved-pool supports appear in Fig. 4b. In the reduced citrate-cycle network, minimal non-negative conservation vectors automatically isolate key metabolic moiety pools, recovering nucleotide pairs, a quinone pair, a ferredoxin pair, coenzyme A, and a lipoyl cofactor pool without manual annotation.

The reduced glycolysis model contains 14 species and 15 directed reactions ($n = 16$, $\ell = 4$, rank $s = 11$, and deficiency $\delta = 1$). Its primary conserved moiety spans eleven carbohydrate intermediates, assigning weight two to hexoses and one to trioses (Fig. 5), capturing exact three-carbon backbone conservation across

aldolase cleavage. Glucose and oxidized ferredoxin form uncovered singleton siphons requiring replenishment, whereas polyphosphate forms a covered siphon because glucokinase retains the polymer on both sides.

From a single initial glucose molecule, the prescribed glycolytic flux yields a certified 15-step pathway realization terminating in two pyruvates, exploring 30 intermediate markings. Aldolase cleavage generates two triose equivalents that feed the repeated lower payoff phase. Step-by-step certificate replay confirms intermediate availability at every enzyme step and verifies exact invariance of the weighted three-carbon backbone pool ($2 \times \text{hexoses} + 1 \times \text{trioses} = 2$). Attempting to execute the lower payoff phase from glucose alone fails because the preparatory intermediates have not yet formed.

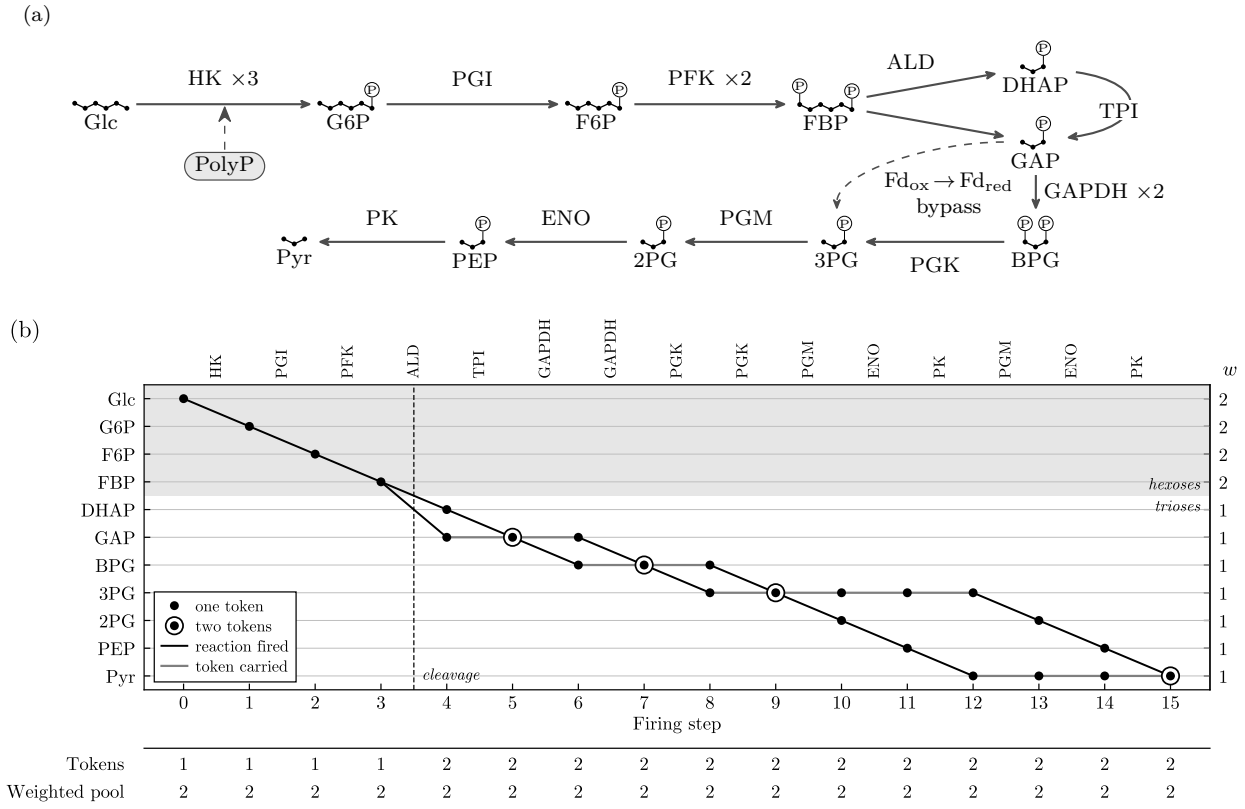


Fig. 5: Conserved pools and pathway realizability in reduced `KEGG` glycolysis (`M00001`). **(a)** Bipartite network retaining polyphosphate and the ferredoxin bypass (dots: carbons, circled P: phosphate). **(b)** Token markings along the certified 15-step conversion of one glucose to two pyruvates (single dot: one token, ringed dot: two tokens). Black lines trace consumed intermediates, gray lines carry unchanged pools, column headers name enzymes, and the dashed line marks aldolase cleavage. Shaded rows indicate hexoses, with pool weights w on the right. Bottom rows trace total carbohydrate tokens and the invariant weighted pool ($2 \times \text{hexoses} + 1 \times \text{trioses} = 2$).

This incidence representation resolves information typically lost under net stoichiometric reduction (Fig. 6). Eliminating shared polyphosphate from the reaction complexes of `R02189` leaves net stoichiometry $\mathbf{S}$, its nullspace, and its conservation laws unchanged ($\mathbf{S}_{\text{retained}} = \mathbf{S}_{\text{cancelled}}$). Yet this reduction alters the reaction topology, reducing complex count from 16 to 14, linkage classes from four to three, and deficiency from one to zero. Net reduction also destroys catalytic gating: the reduced reaction fires freely without polyphosphate, whereas explicit incidence blocks firing until polyphosphate is supplied. Explicit reactant-product incidences thus enforce physical cofactor requirements discarded by net matrices.

Restoring currency metabolites (ATP, ADP, NAD^+, NADH, P_i, and PP_i) yields the complete 24-species metabolic network. Supplied solely with glucose, hexokinase cannot fire due to ATP depletion, and exhaustive search certifies that the pathway is unrealizable. The reduced model establishes topological feasibility of intermediate conversions, whereas full pathway execution requires balancing cofactor regeneration and energy charge (detailed with inventory controls in Supplementary Section S5).

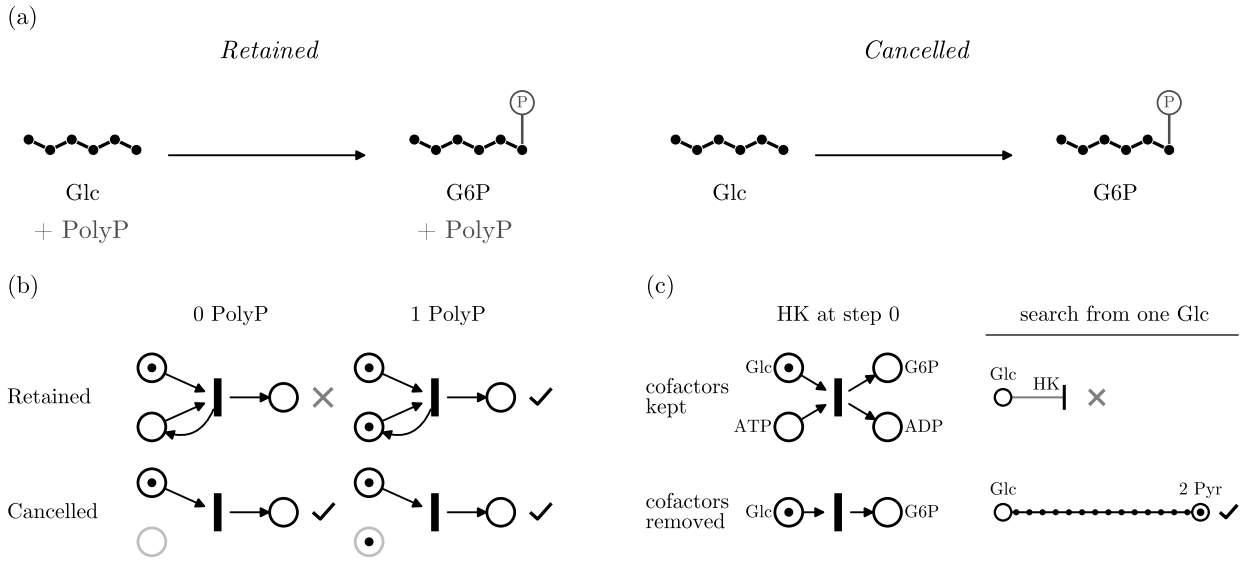


Fig. 6: Sensitivity of structural invariants and pathway realizability to participant retention and currency omission. **(a)** Reaction `R02189` with shared polyphosphate retained versus eliminated in net stoichiometry. **(b)** Petri-net enabling under zero versus one polyphosphate token (check: enabled, cross: blocked). **(c)** Prescribed glycolytic flux from one glucose under retained versus removed currency metabolites. Left: hexokinase (`R01786`) at step 0. Right: pathway reachability from initial glucose (open circle) through intermediate steps (dots) to two pyruvates (ringed dot).

3.5 Generative chemistry: carbon conservation and precursor requirements in formose

The formose reaction, the classic prebiotic condensation of formaldehyde into diverse sugars [20], provides an open-ended, rule-generated benchmark complementary to curated metabolic networks. Seeded with formaldehyde (C_1) and glycolaldehyde (C_2), four core transformation rules (aldol addition, retro-aldol cleavage, and reversible keto-enol tautomerization through an enediol intermediate) iteratively construct the reaction space. After four expansion rounds, the network comprises 37 species and 44 directed reactions ($n = 54$ complexes, $\ell = 17$ linkage classes, rank $s = 36$, and deficiency $\delta = 1$), reaching decose (C_{10}) species. Median cumulative generation time with the native `SynReactor` DPO controller is 0.0177 s across 20 timed runs after three warm-ups.

Matched expansions in `MØD` 1.0.0.75 [4] and `MolFoundry` 0.1.11 [35] independently validate network generation across six expansion depths (Fig. 7a). At depth six, all three tools generate identical sets of 10 572 species and 11 239 stoichiometric reactions. Median cumulative generation times on an isolated logical CPU are 1.36 s for native `SynReactor` DPO, 1.57 s for `MØD`, and 11.8 s for `MolFoundry` (Fig. 7b

and Table S7). These end-to-end API measurements encompass rule parsing, frontier updates, and Python graph construction for `SynKit` without isolating native kernel execution.

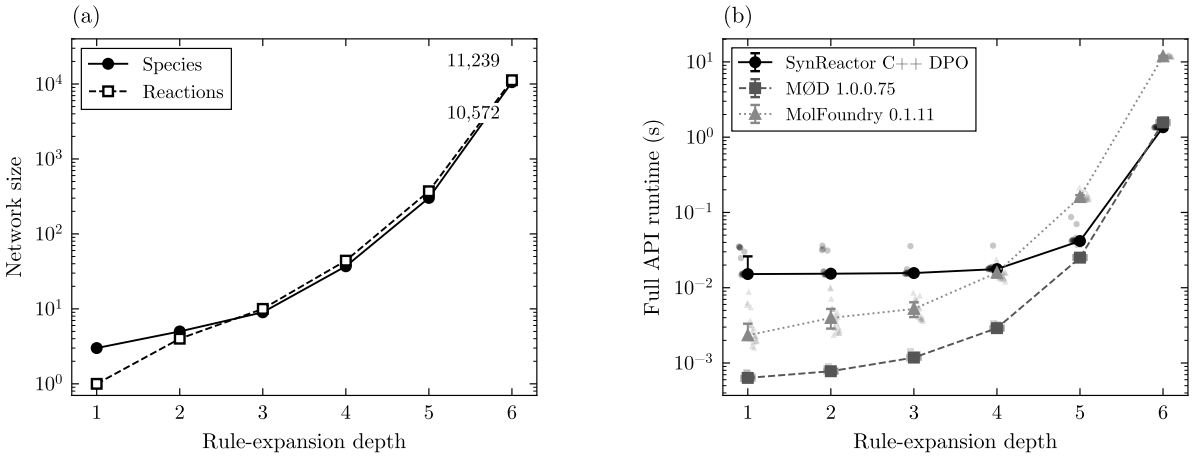


Fig. 7: Formose network growth and full API runtime. **(a)** Species and directed stoichiometric reactions across six rule-expansion depths, with all three tools generating identical sets. **(b)** Full API runtime of native `SynReactor` DPO, `MØD`, and `MolFoundry`. Faint points show 20 timed runs per tool and depth after three warm-ups, large markers show medians, and error bars span the first and third quartiles. Points are offset horizontally for visibility, and times include per-call rule and seed preparation and network construction without imports or engine warm-up. Both panels use logarithmic vertical axes.

Exact nullspace analysis of the four-iteration stoichiometric matrix reveals a one-dimensional left nullspace. Its positive primitive conservation vector agrees identically with the heavy-atom carbon count of every generated molecule, confirming that automated graph transformation preserves carbon conservation across all 37 species (Fig. 8b).

Qualitative hypergraph search indicates that representative aldotetrose (T_4) and aldohexose (T_6) targets are reachable from the seed pair, but discrete Petri-net realizability differentiates precursor pools sharing identical carbon counts (Fig. 8c). Supplying six carbon atoms as four formaldehydes and one glycolaldehyde (M_3) fails to realize either route. Supplying two formaldehydes and two glycolaldehydes (M_4) supports T_4 synthesis, while three glycolaldehydes without formaldehyde (M_5) realizes both routes. Target synthesis thus hinges on precursor structural assembly and intermediate availability rather than total elemental mass. All ten route–inventory decisions match independent evaluation in `MolFoundry`, and all positive certificates pass independent step-by-step replay (pathway mechanisms and pools detailed in Supplementary Section S6).

4 Conclusions

The CRN submodule of `SynKit` unites stoichiometric conservation, chemical reaction network theory, and pathway realizability on a bipartite representation preserving distinct reactant and product incidences. Across curated metabolic modules and generative chemistries, these analyses demonstrate that net stoichiometric reduction obscures essential physical constraints. Retaining shared participants preserves network

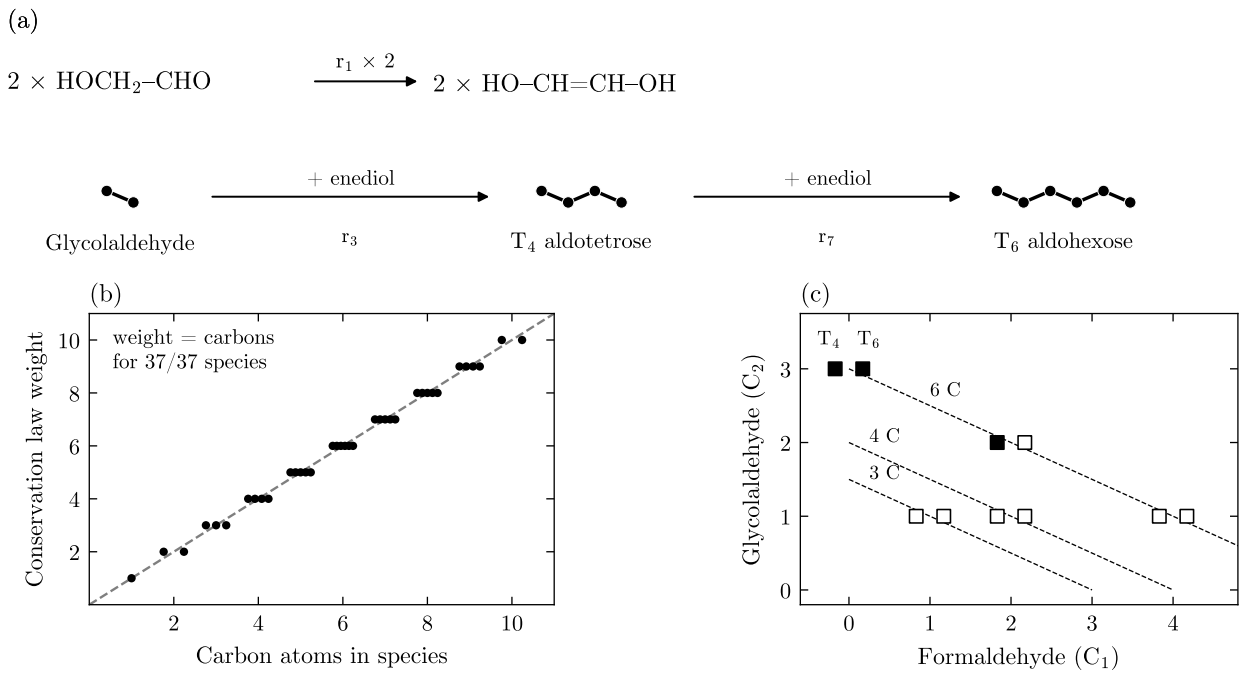


Fig. 8: Carbon conservation and precursor-dependent pathway realizability in the generated formose network. **(a)** Target hexose route requiring three glycolaldehyde precursors via two enolizations (r_1) and two aldol additions (r_3, r_7), each addition consuming one enediol. **(b)** Left-nullspace conservation weights against carbon atom counts across all 37 species (points jittered horizontally, dashed line: $y = x$). **(c)** Five initial inventories by formaldehyde and glycolaldehyde count, with dashed lines joining inventories of equal total carbon. The left and right squares of each inventory refer to the tetrose (T_4) and hexose (T_6) routes, filled when a certificate is verified and open when exhaustive search fails. Inventories with six carbon atoms differ in which routes they can supply.

deficiency and enforces catalytic gating, preventing reactions from firing without required cofactors. Similarly, isolating currency cofactors reveals carbon backbone realizability while showing that full pathway execution requires cofactor replenishment. In generative networks, exact mass conservation is necessary but insufficient for pathway synthesis, where precursor stoichiometric ratios and assembly motifs govern realizability from finite pools. By employing exact rational arithmetic and replayable certificates, `SynKit` makes these modeling assumptions explicit, reproducible, and testable.

These analyses characterize structural enabling and discrete token sufficiency rather than continuous reaction rates, concentrations, thermodynamic driving forces, or *in vivo* regulatory kinetics. Reachability certificates evaluate prescribed reaction multisets rather than unrestricted *de novo* route synthesis, and exact siphon and semiflow enumeration faces combinatorial scaling on dense networks. Structural `SBML` exchange is confined to Core stoichiometric topology, excluding kinetic rate laws and non-core extensions.

Future work will couple discrete firing certificates with estimated reaction free energies ($\Delta_r G$) to filter generated reaction spaces thermodynamically before kinetic simulation. Integrating `SynKit`'s incidence representation with constraint-based flux methods and dynamic ODE simulators will bridge topological feasibility with quantitative metabolic phenotypes, while reducing cross-boundary serialization will scale exploration across complex prebiotic reaction spaces.

Availability and requirements

Project name: `SynKit` (CRN submodule)

Project home page: github.com/TieuLongPhan/SynKit

Software version: `SynKit` 1.6.5 [40]

Operating system(s): Platform independent (Linux, macOS, Windows)

Programming language: Python, C++17, and C++20

Other requirements: Python $\geq$ 3.10, `NetworkX` $\geq$ 3.3, `NumPy` [41], `SciPy` [42], `RDKit`, Boost headers, CMake

License: MIT

Any restrictions to use by non-academics: None

List of abbreviations

ACR: Absolute concentration robustness, **CRN**: Chemical reaction network, **CRNT**: Chemical reaction network theory, **DPO**: Double pushout, **EMP**: Embden–Meyerhof–Parnas, **FBA**: Flux balance analysis, **KEGG**: Kyoto Encyclopedia of Genes and Genomes, **SBML**: Systems Biology Markup Language, **SMARTS**: SMILES arbitrary target specification, **TCA**: Tricarboxylic acid, **WR**: Weakly reversible.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The CRN submodule is distributed with `SynKit` under the MIT License. Source code is available on GitHub at github.com/TieuLongPhan/SynKit [40], and the reproducible analysis scripts are located in the `Experiment/CRN` directory. `KEGG` [1] pathways remain subject to `KEGG` terms, and `BioModels` entries are distributed under the CC0 1.0 Public Domain Dedication.

Use of generative AI

Claude Opus (Anthropic) was used for language editing and TikZ figure formatting. The authors reviewed, verified, and take full responsibility for all manuscript content.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

Author contributions are reported using the CRediT taxonomy. **T.-A.P.**: Software, Validation, Writing (original draft), Writing (review and editing). **T.-L.P.**: Conceptualization, Methodology, Software, Validation, Writing (original draft), Writing (review and editing). Both authors read and approved the final manuscript.

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